

Daniel Dubinko

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Education

Queen's University

Bachelor of Applied Science, Mechatronics and Robotics Engineering

- First Class Honours, Dean's Scholar | GPA: 4.02/4.3 (Last Two Years)

Kingston, ON

Sep. 2020 – May 2026

Experience

Guidance Navigation & Controls Engineer Co-op

MDA Space

Jan. 2025 – Aug. 2025

Brampton, ON

- Built **kinematic bicycle-model** controls and **Dubins-curve** path optimization for an **RRT*** planner, improving feasible state-space connections and reaching **95% goal-point reachability**.
- Created a reusable **Pandas** library for ML data augmentation, adopted by the AI and Path Planning teams, **cutting data-preparation time by at least 50%**.
- Implemented unit tests using Python to ensure system coverage, leveraging frameworks like **pytest** for automated testing.

Autonomy Team Intern

MacLean Engineering

Jun. 2024 – Dec. 2024

Collingwood, ON

- Built a **C++ ROS 2** teleoperation node for an underground mining vehicle, tested end to end in **Gazebo**.
- Implemented an **Extended Kalman Filter** to fuse IMU and odometry measurements, **cutting dead-reckoning drift by 83%**.
- Deployed **Docker** containers with **GitLab CI/CD pipelines** in an Agile project management environment.

Controls Team Developer

Queen's Knights Robotics Team (RoboMaster)

Sep. 2025 – Apr. 2026

Kingston, ON

- Designed a **C++** flywheel class supporting **CAN-** and **GPIO-driven** flywheels, switchable through a single state variable, replacing manual code commenting between setups.
- Developed dual-mode **C++** control (keyboard/mouse + RC) with an **RC safety override** that lets the operator disable keyboard input and take direct control.

Undergraduate Research Assistant

Ingenuity Labs, Queen's University

May 2023 – Aug. 2023

Kingston, ON

- Built a VR system for real-time human-robot high-fives, localizing a tracked user's hand and a **KUKA** arm end effector in one shared frame by integrating **Vicon** motion capture with **Unity**.
- Implemented **Jacobian-based inverse kinematics** to map the user's hand pose (position and quaternion orientation) to KUKA arm waypoints, achieving a **70% hand-contact success rate**.

Projects

Autonomous Litter-Collecting Quadruped Robot

- Designed a modular **2D LiDAR** and **IMU** localization unit with a vibration-damped IMU mount that attaches to any robot to provide odometry; deployed it on a Unitree Go1.
- Integrated **SLAM** and **Nav2** to generate waypoint commands sent over the Unitree SDK, **achieving 90% waypoint success**.
- Built a **ROS 2 Jazzy** state machine for search, approach, and grab behaviour, sending state changes over **UDP** with **Protocol Buffers**.

FlipWalker Robot State Estimation

- Designed **closed-loop gait control** using a 3-state FSM, replacing open-loop timed servo control of a flipping robot.
- Developed **ESP32 C++** firmware that estimates joint angle with magnetometers, using a 7-point LUT across a $\pm 90^\circ$ sweep.

Artificial Potential Field Path Planner

- Implemented a real-time **potential-field planner** using a conical potential while traversing obstacles and a parabolic potential to converge on the goal, with local-minima handling and obstacle safety thresholds.

Autonomous Mobile CO₂ Monitoring Robot

- Integrated **ToF** and **LiDAR** sensors with **ROS Noetic** for **SLAM**, enabling continuous live mapping of indoor environments.
- Designed and integrated a custom electronics enclosure using **Onshape** for the mobile robot.
- Improved robot stability and control performance through **PID tuning** and control system techniques.

Technical Skills

Languages and Frameworks: C++, Python, MATLAB, Simulink, ROS 2, Unitree SDK

Systems & Integration: UDP/TCP, Protocol Buffers, CAN

Libraries: NumPy, Pandas, OpenCV, pytest, Matplotlib

Developer Tools: Git, Docker, Jira, Confluence